Physics — Grade 6 Formula Sheet

SI Base Units

Quantity	Unit	Symbol
length / distance	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

Metric Prefixes

Power	Prefix	Power	Prefix
10^{12}	T tera	10^{-1}	d deci
10^{9}	G giga	10^{-2}	c centi
10^{6}	M mega	10^{-3}	m milli
10^{3}	k kilo	10^{-6}	μ micro
10^{2}	h hecto	10^{-9}	n nano
10^{1}	da deka	10^{-12}	p pico

- No prefix $= 10^{0} = 1$ (the base unit).

Scientific Notation

$$a \times 10^{n}$$

- **Big** number → decimal moves **left** → **positive** n.
- **Small** number → decimal moves **right** → **negative** n.
- A prefix is just a power of ten: kilo $= 10^{3}$, nano $= 10^{-9}$.

$9{,}300{,}000 = 9.3 \times 10^{6}$
$0.0000045 = 4.5 \times 10^{-6}$

Unit Conversion

Multiply by fractions equal to 1; cancel unwanted units.

$$5\ \cancel{\mathrm{km}}\left(\frac{1000\ \mathrm{m}}{1\ \cancel{\mathrm{km}}}\right) = 5000\ \mathrm{m}$$

Speed units

$\frac{\mathrm{km}}{\mathrm{h}} \div 3.6 = \frac{\mathrm{m}}{\mathrm{s}}$ $\quad$ $\frac{\mathrm{m}}{\mathrm{s}} \times 3.6 = \frac{\mathrm{km}}{\mathrm{h}}$

$40\ \frac{\mathrm{km}}{\mathrm{h}} = 40 \times \frac{1000}{3600} = 11.1\ \frac{\mathrm{m}}{\mathrm{s}}$

Common: $1\ \mathrm{km} = 1000\ \mathrm{m} \cdot 1\ \mathrm{h} = 3600\ \mathrm{s} \cdot 1\ \mathrm{min} = 60\ \mathrm{s} \cdot 1\ \mathrm{kg} = 1000\ \mathrm{g}$

Derived & Special Units

Quantity	Formula	Base units	Name
Force	$F = ma$	$\mathrm{kg \cdot m/s^2}$	Newton (N)
Energy / Work	$W = Fd$	$\mathrm{kg \cdot m^2/s^2}$	Joule (J)
Power	$P = W/t$	$\mathrm{kg \cdot m^2/s^3}$	Watt (W)
Frequency	$f = 1/T$	$\mathrm{1/s}$	Hertz (Hz)
Charge	$Q = It$	$\mathrm{A \cdot s}$	Coulomb (C)

Symbols & Definitions

Quantity	Symbol	Unit	S/V
position	x	m	V
distance	d	m	S
displacement	Δx	m	V
time	t	s	S
speed	s	$\mathrm{m/s}$	S
velocity	v	$\mathrm{m/s}$	V
acceleration	a	$\mathrm{m/s^2}$	V
mass	m	kg	S
force	F	N	V
net force	F_{net}	N	V
weight	W	N	V
friction	f	N	V
gravity	g	$9.8\ \mathrm{m/s^2}$	V

S = scalar, V = vector.

Scalars & Vectors

- **Scalar** — magnitude only (mass, time, distance, speed).
- **Vector** — magnitude **and** direction (position, displacement, velocity, acceleration, force).
- Collinear vectors (same axis): **same** direction → **add**; **opposite** → **subtract**.

Same Direction $\xrightarrow{5\ \mathrm{m}} + \xrightarrow{3\ \mathrm{m}} = \xrightarrow{8\ \mathrm{m}}$

Opposite Direction $\xrightarrow{5\ \mathrm{m}} + \xleftarrow{3\ \mathrm{m}} = \xrightarrow{2\ \mathrm{m}}$

Direction	Sign
right / east / up / north	$+$
left / west / down / south	$-$

Kinematics

- Distance d = total path length (scalar). Displacement Δx = change in position (vector).
- $\Delta x = x - x_0$ (x final, x_0 initial position)
- Speed: $s = \dfrac{d}{t}$
- Velocity: $v = \dfrac{\Delta x}{t}$
- Average speed $= \dfrac{d_{\text{total}}}{t_{\text{total}}}$

  Average velocity $= \dfrac{\Delta x_{\text{total}}}{t_{\text{total}}}$
- Acceleration: $a = \dfrac{\Delta v}{t} = \dfrac{v - v_0}{t}$

Position–time graph

The **slope** of a position–time graph is the **velocity**. Positive slope → $+v$; negative slope → $-v$; flat line → at rest.

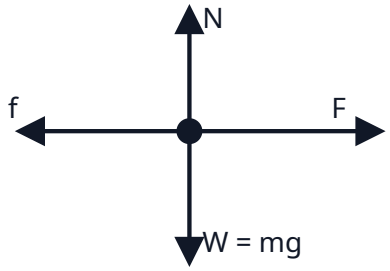
Forces

- **Newton's 1st law:** an object keeps its motion unless a **net force** acts.
- **Newton's 2nd law:** $F_{\text{net}} = ma$ (acceleration is in the direction of the net force).
- **Newton's 3rd law:** for every action there is an equal and opposite reaction (on two different objects).
- **Net force:** $F_{\text{net}} = \sum F$ (add same direction, subtract opposite).
- **Weight:** $W = mg$, with $g = 9.8\ \text{m/s}^2$ on Earth.
- **Equilibrium:** $F_{\text{net}} = 0 \Rightarrow a = 0 \Rightarrow$ constant velocity (may be zero — at rest).
- **Drag** increases with speed; at **terminal velocity** drag = weight, so $F_{\text{net}} = 0$.

$$1\ \text{N} = 1\ \text{kg} \cdot \frac{\text{m}}{\text{s}^2}$$

Free-Body Diagrams

Draw every force acting on **one** object as a labeled arrow from a dot. Include only forces **on** the object; arrow length = magnitude, head = direction.



Simple Machines

A machine changes the **size** or **direction** of a force (not the total work). **Lever · Wheel & axle · Pulley · Wedge · Screw · Inclined plane.**

Problem-Solving Steps

1. Annotate: what is given? what is the unknown?
2. What equation connects the quantities?
3. Plug in with units and solve.
4. Check the units and direction.

Hints

"Starts from rest" → $v_0 = 0$.
"Comes to rest" → $v = 0$. Give a direction for every vector quantity.